
CFD Modeling of a close loop enhanced geothermal system



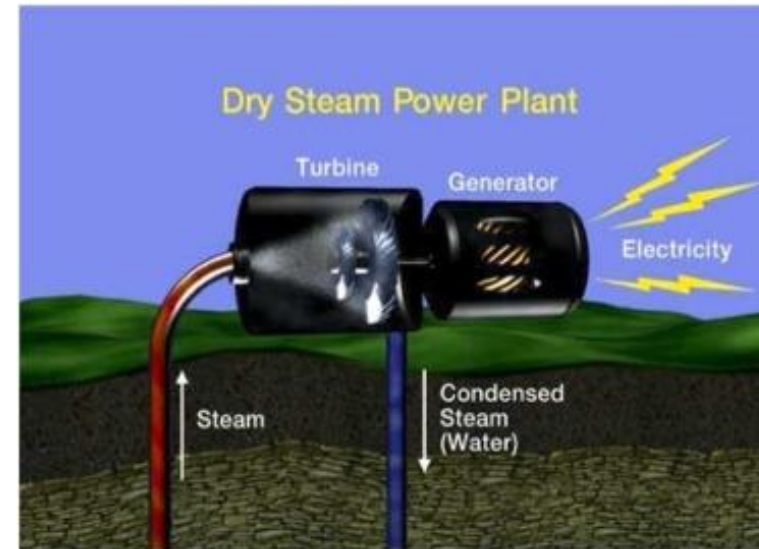
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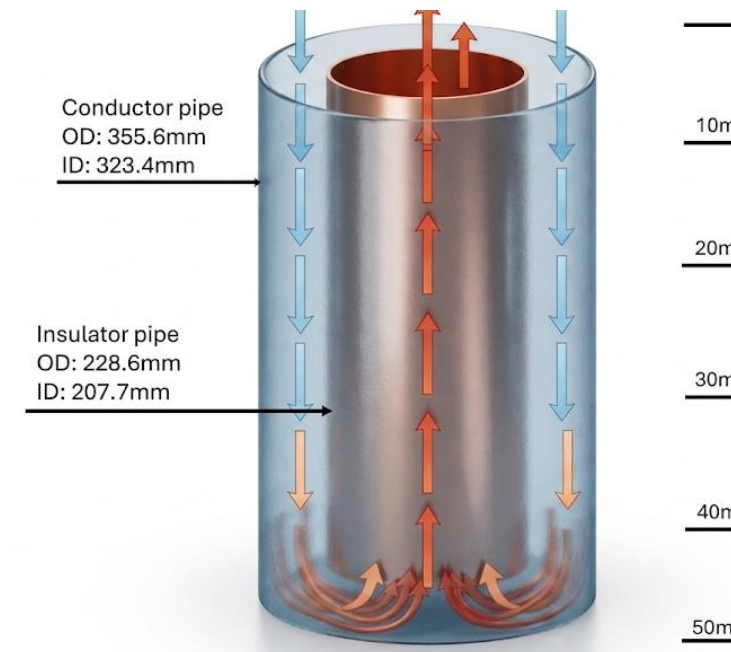
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Motivation:

- **High cost** of conventional geothermal systems (multiple-wells).
- **Limited applicability** to high-temperature reservoirs only.
- **Environment concerns** from fluid extraction and re-injection.
- **Need for a cost-effective, sustainable** solution (closed loop system).



Abdulrazzaq et al. (2026)



Literature review:

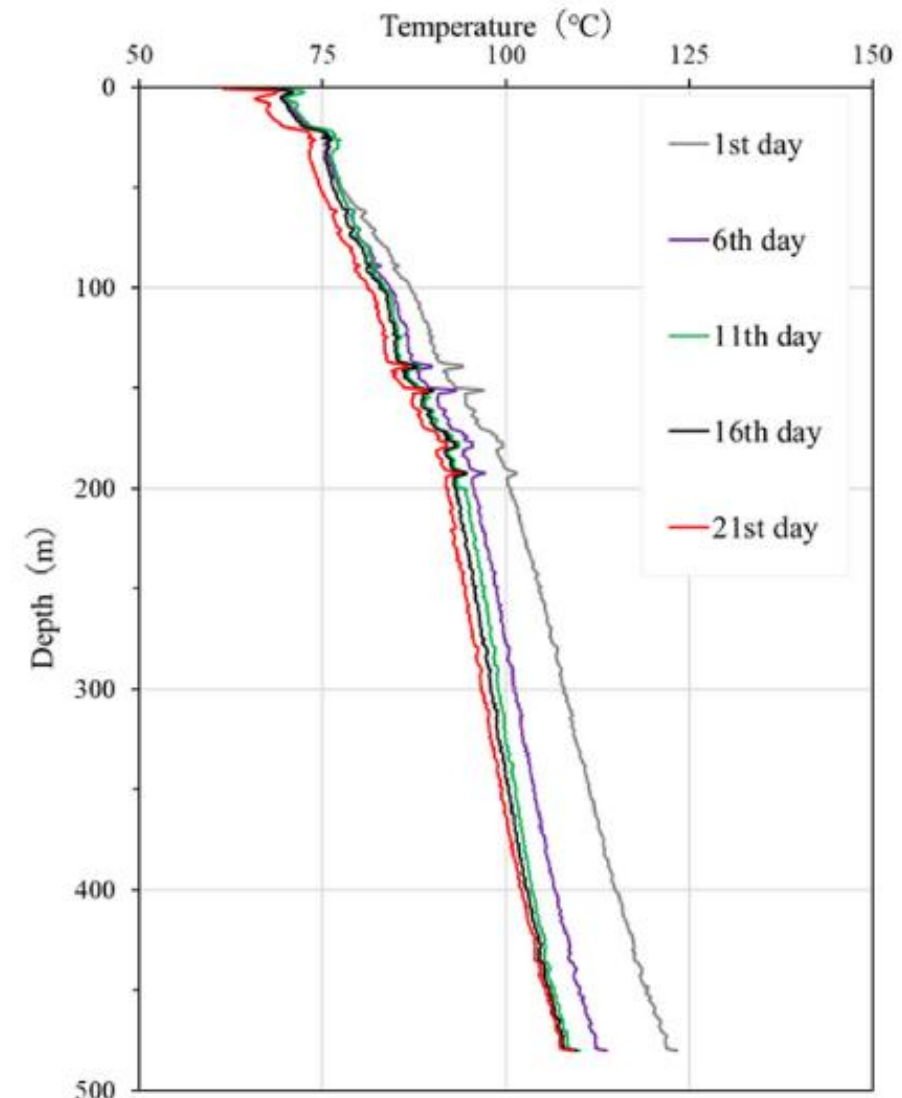


Pokhrel et al. (2022) – Field Experiment

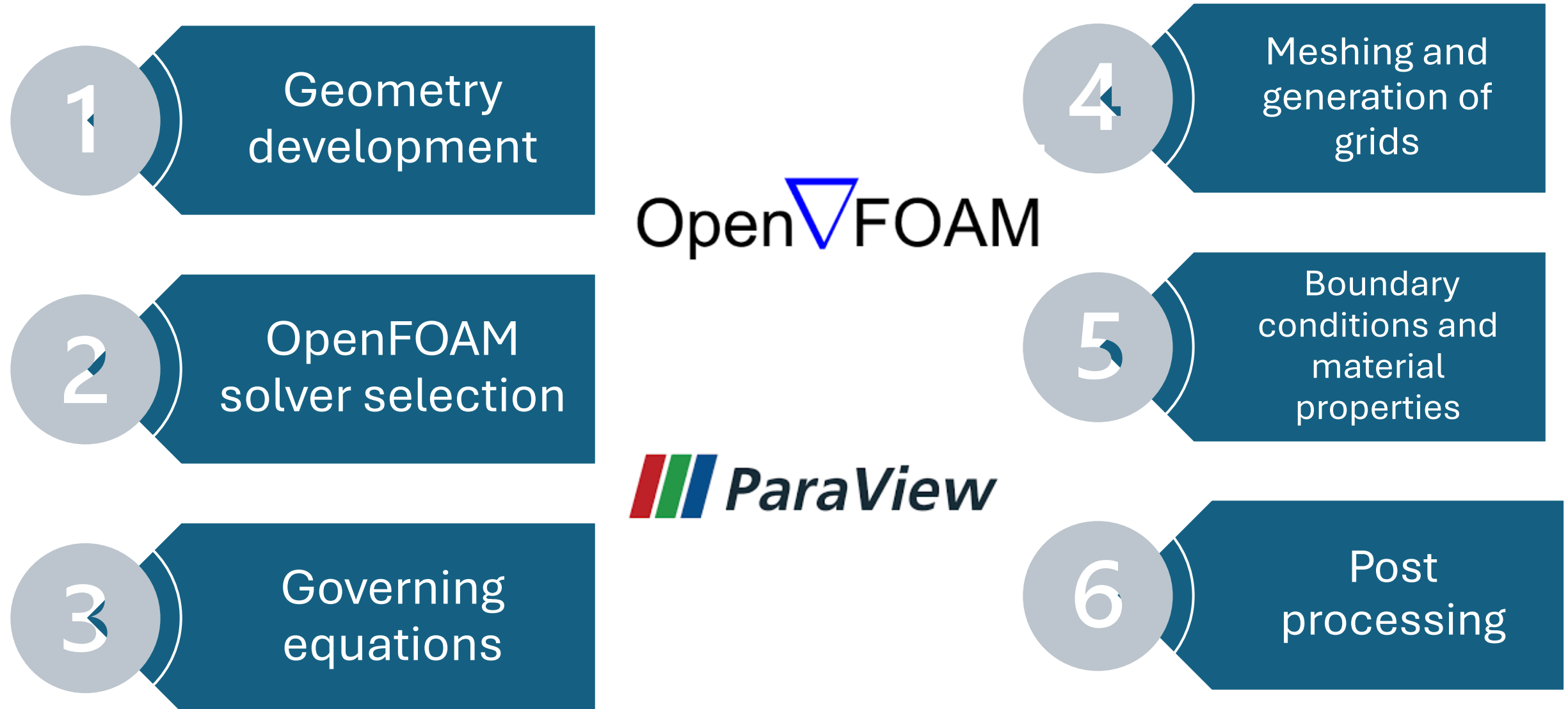
The temperature profiles show progressive thermal depletion along the 500 m borehole over 21 days of operation.

Gap Addressed by Present Study

Prior transient model (Pokhrel et al.) did not evaluate **net useful energy**. This work adds steady-state CFD analysis and subtracts **pumping losses** to quantify true efficiency.



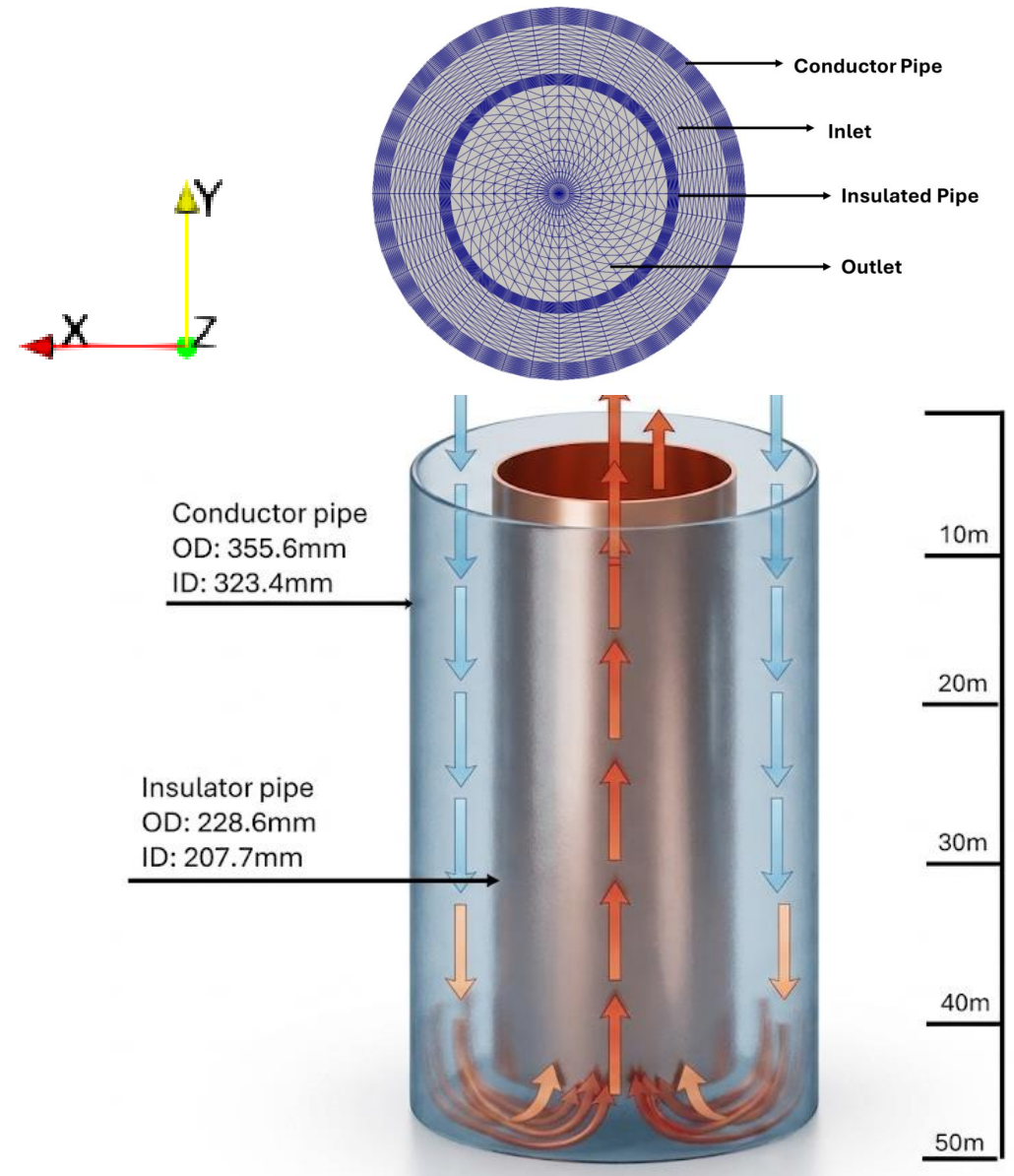
Methodology:



Geometry:

Structure parameter	Conductor Pipe	Insulated Pipe
Inner diameter (mm)	323.4	207.7
Outer diameter (mm)	355.6	228.6
Depth (m)	50	49

S. Pokhrel et al. (2022)



Solver selection:



The following assumption are invoked:

- The flow is steady-state
- The working fluid is incompressible
- Gravitational body forces are neglected
- Thermophysical properties (such as μ , C_p and k) are constant
- The OpenFOAM solver we used is **chtMultiRegionSimpleFoam**

Governing Equations:



Steady state Incompressible Continuity Equation: $\nabla \cdot \vec{v} = 0$

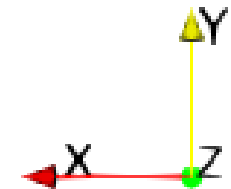
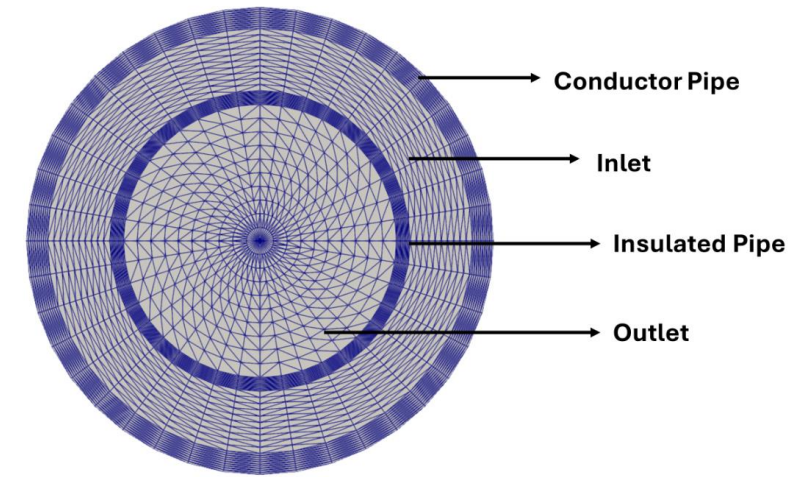
Steady state Momentum Equation: $\rho(\vec{v} \cdot \nabla \vec{v}) = -\nabla p + \mu \nabla^2 \vec{v}$

Steady state Energy Equation: $\rho C_p \vec{v} \cdot \nabla T = k \nabla^2 T$

Meshing and Grid generation:



Parameters	Description
Utility	blockMesh
Cell type	Structured hexahedral
Total cell	1.92 million
Radial cell	Resolution 10 per block
Circumferential cell	Resolution 10 per block
Axial cell	Resolution 1200

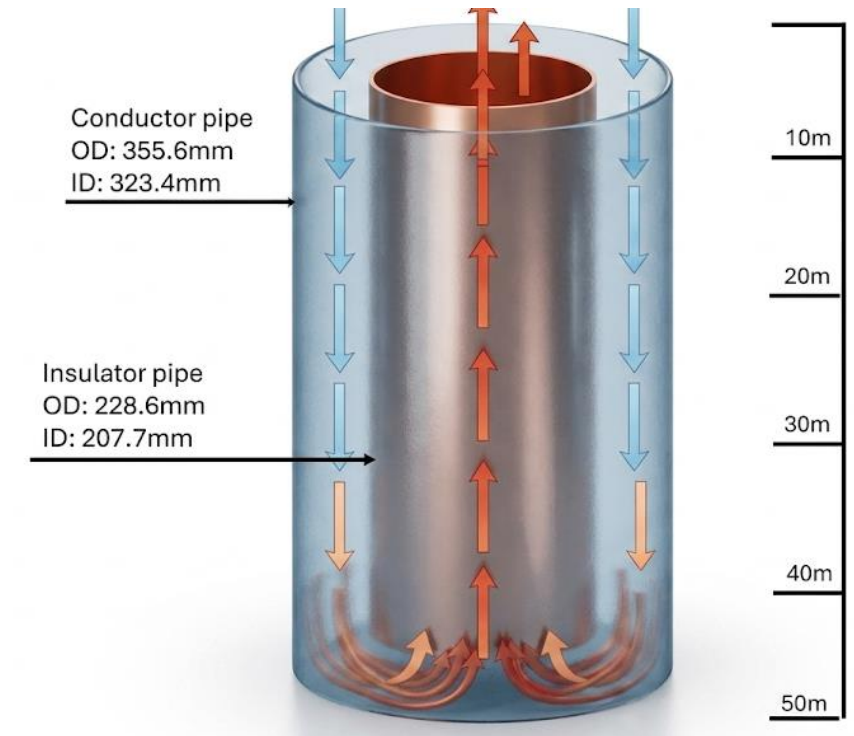


Boundary conditions:



Temperature

- Working fluid Inlet: Dirichlet condition: $T_{\text{inlet}} = 343 \text{ K}$
- Conductor pipe: $T_{\text{wall}}(z) = 343 + 0.38z \text{ (K)}$
- Bottom face at $z = 50\text{m}$: $T_{\text{bottom}} = 362 \text{ K}$
- Working fluid Outlet: inletOutlet condition
- Fluid-solid interface:
compressible::turbulentTemperatureRadCoupledMixed
 $T_{\text{solid}} = T_{\text{fluid}}$ and $\left(k \frac{dT}{dr}\right)_{\text{solid}} = \left(k \frac{dT}{dr}\right)_{\text{fluid}}$





Velocity

- Inlet (annulus top face) : Dirichlet condition: $U_{\text{inlet}} = (0,0, V) \text{ m/s}$ where, $V = m/(\rho.A)$
- Outlet (inner pipe top face): inletOutlet condition
- All solid walls and fluid-solid interfaces: no-slip condition: $U_{\text{wall}} = (0,0,0) \text{ m/s}$

Pressure

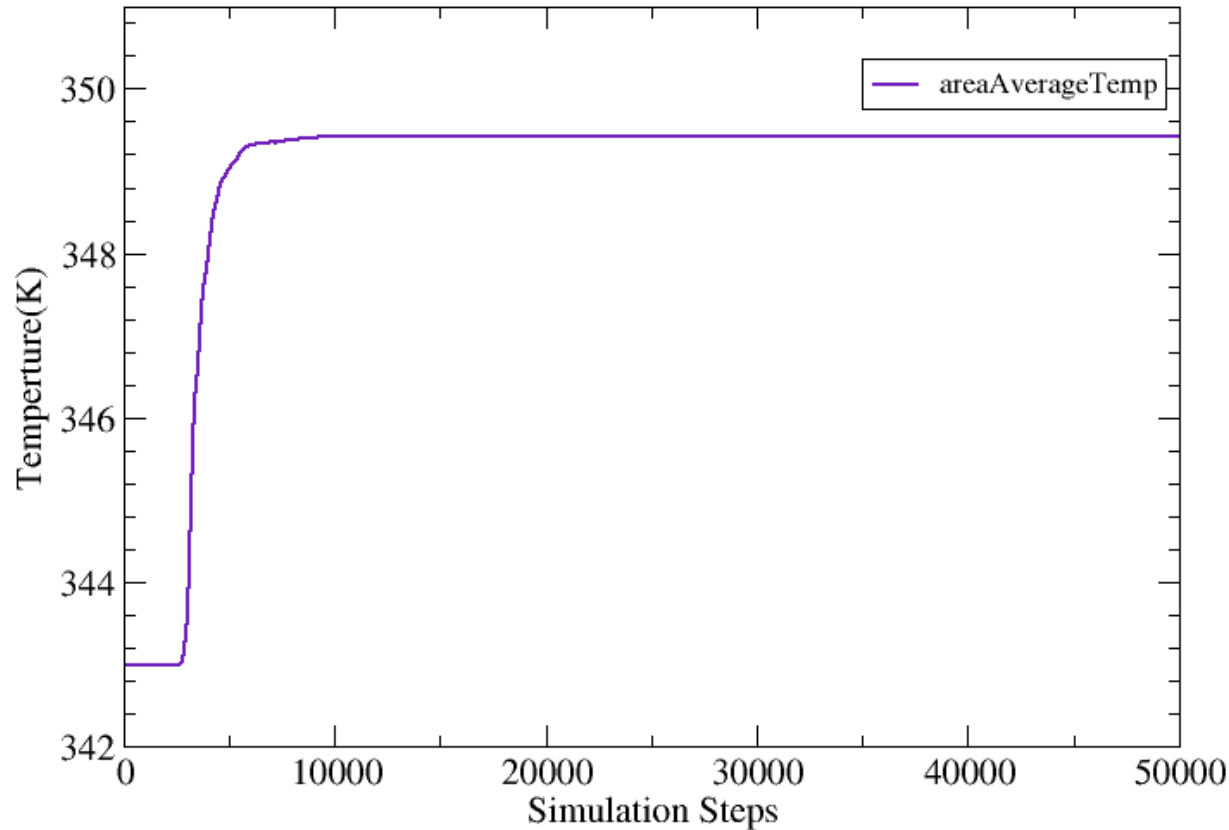
- The SIMPLE – Semi-implicit method for pressure linked equation

Material properties:

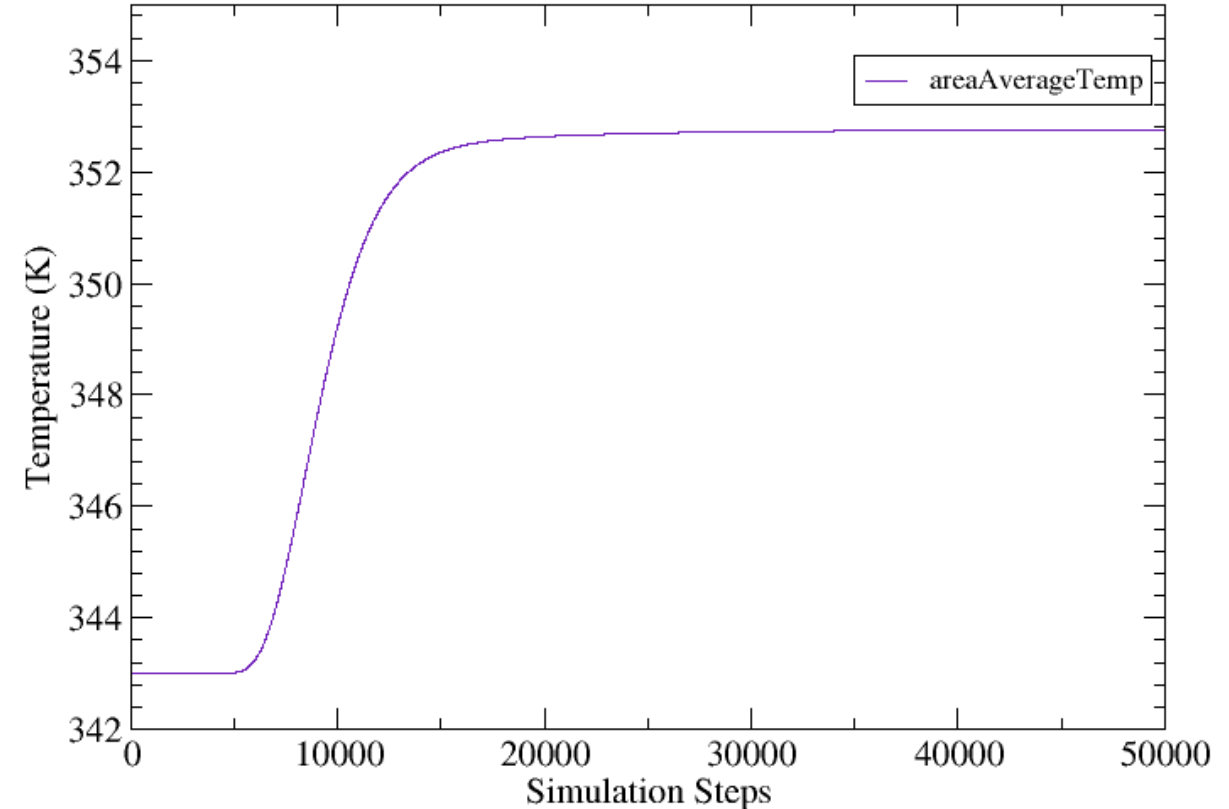


Materials	Density $\left(\frac{kg}{m^3}\right)$	Specific heat $\left(\frac{J}{kg K}\right)$	Thermal Conductivity $\left(\frac{W}{m K}\right)$
Water	998.2	4182	0.6
Insulated inner pipe (carbon steel +foam rubber)	5240	310	0.05
Conductor pipe	7850	470	46.1

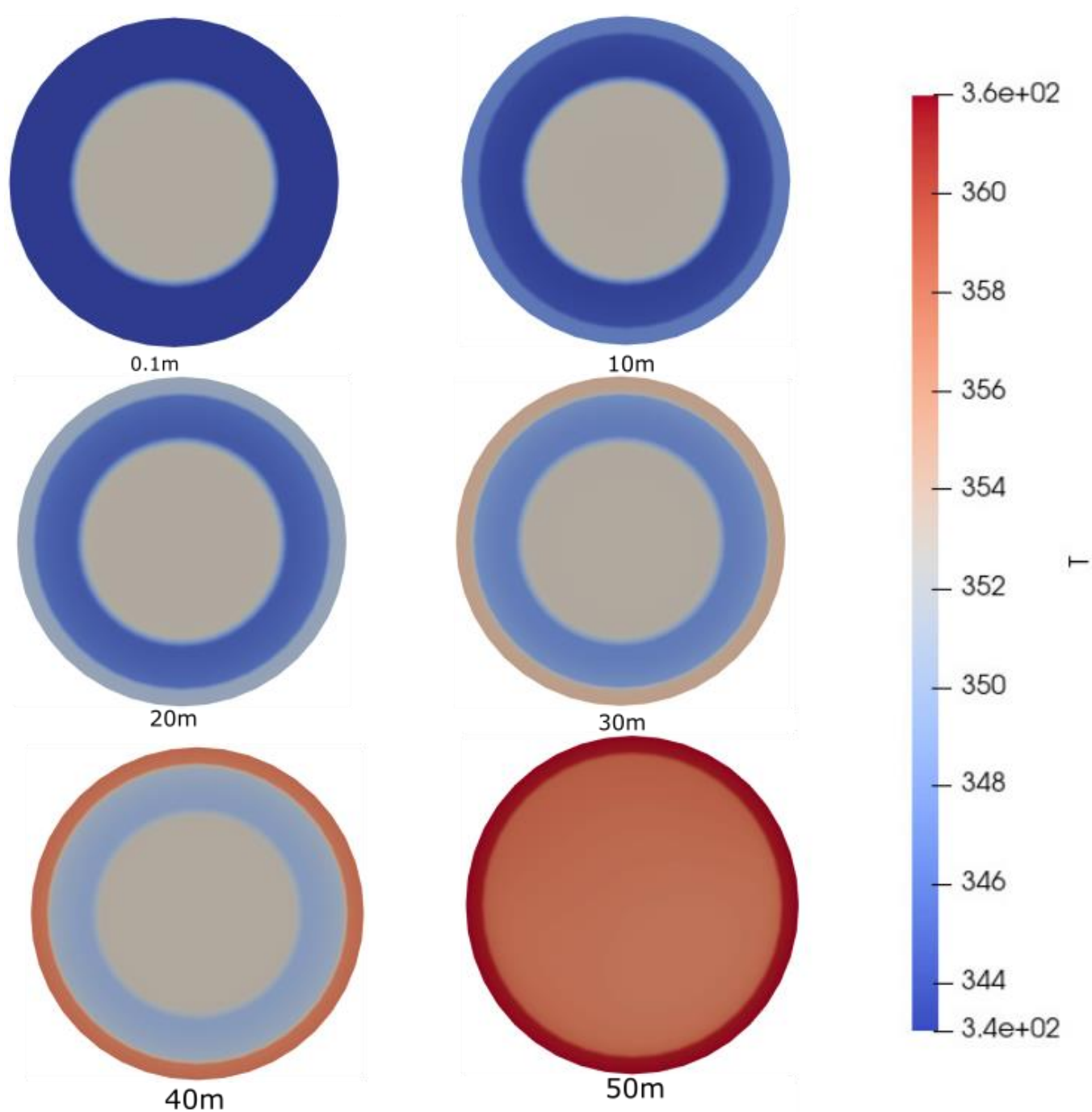
Steady-State Convergence:



Laminar regime

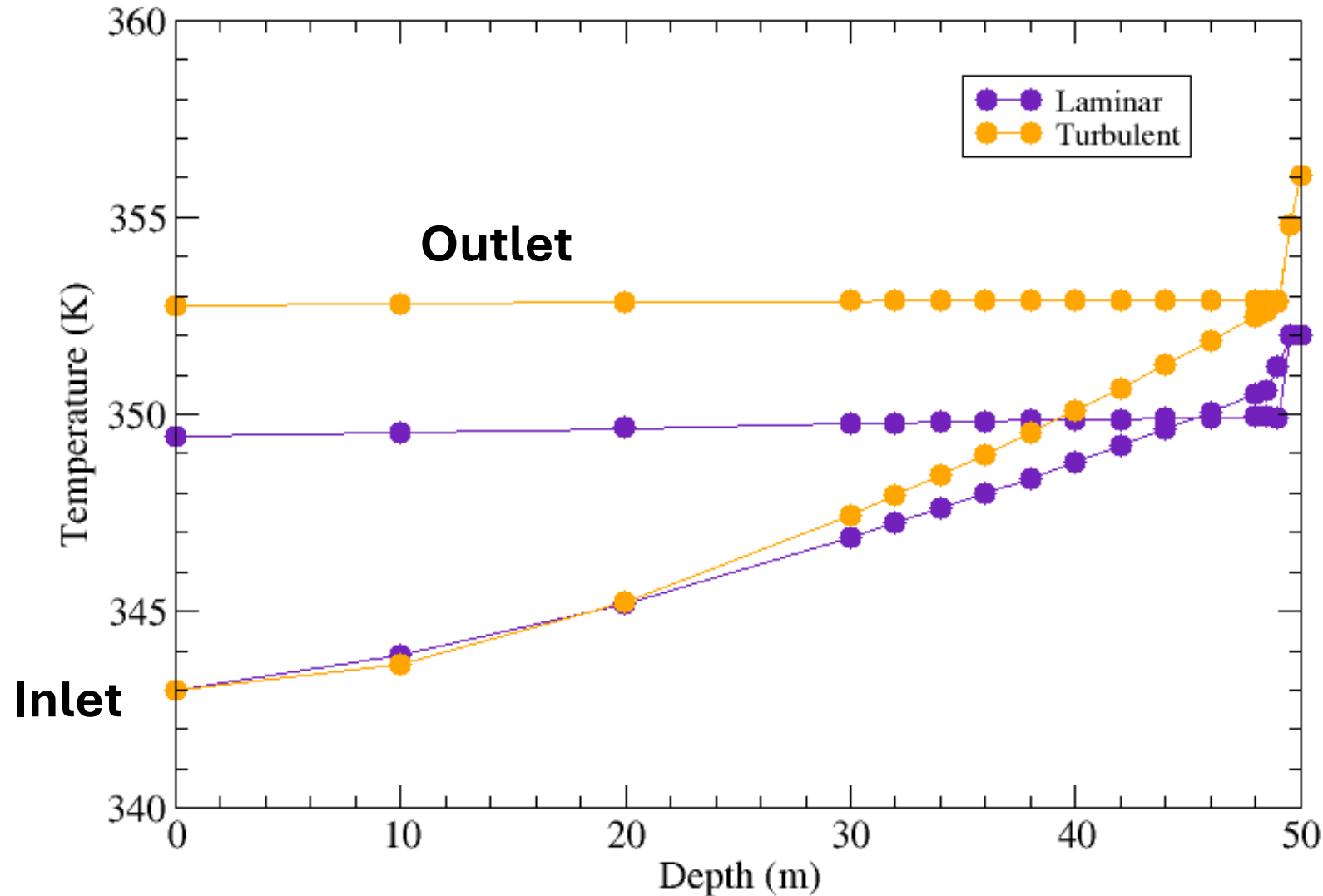


Turbulent regime



The cross-sectional top view provides the visual representation of thermal development within the system along the depth from the inlet $z=0.1$ m to the bottom $z=50$ m

Temperature Profiles:

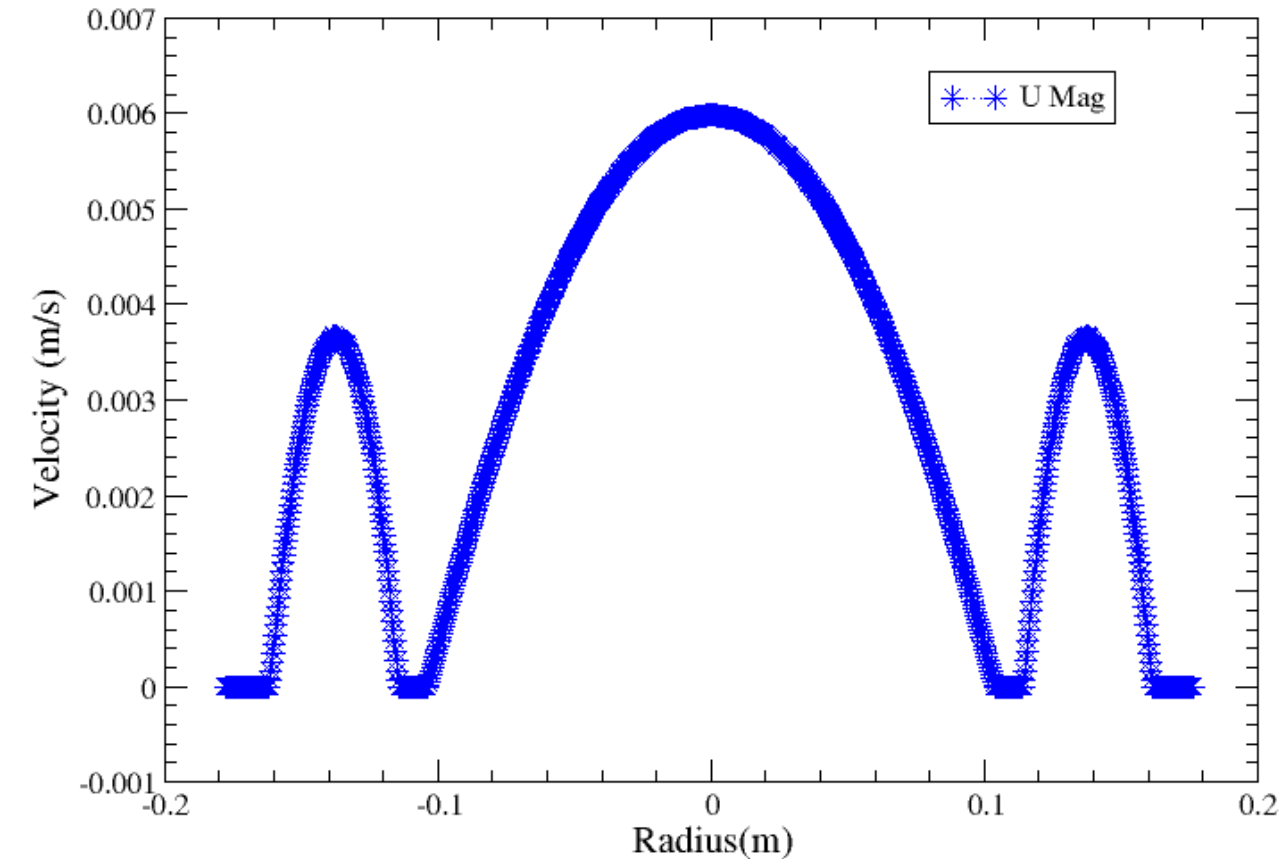


At $z = 50\text{m}$: $T_{\text{bottom}} = 362\text{ K}$

Velocity profiles(fully developed region):

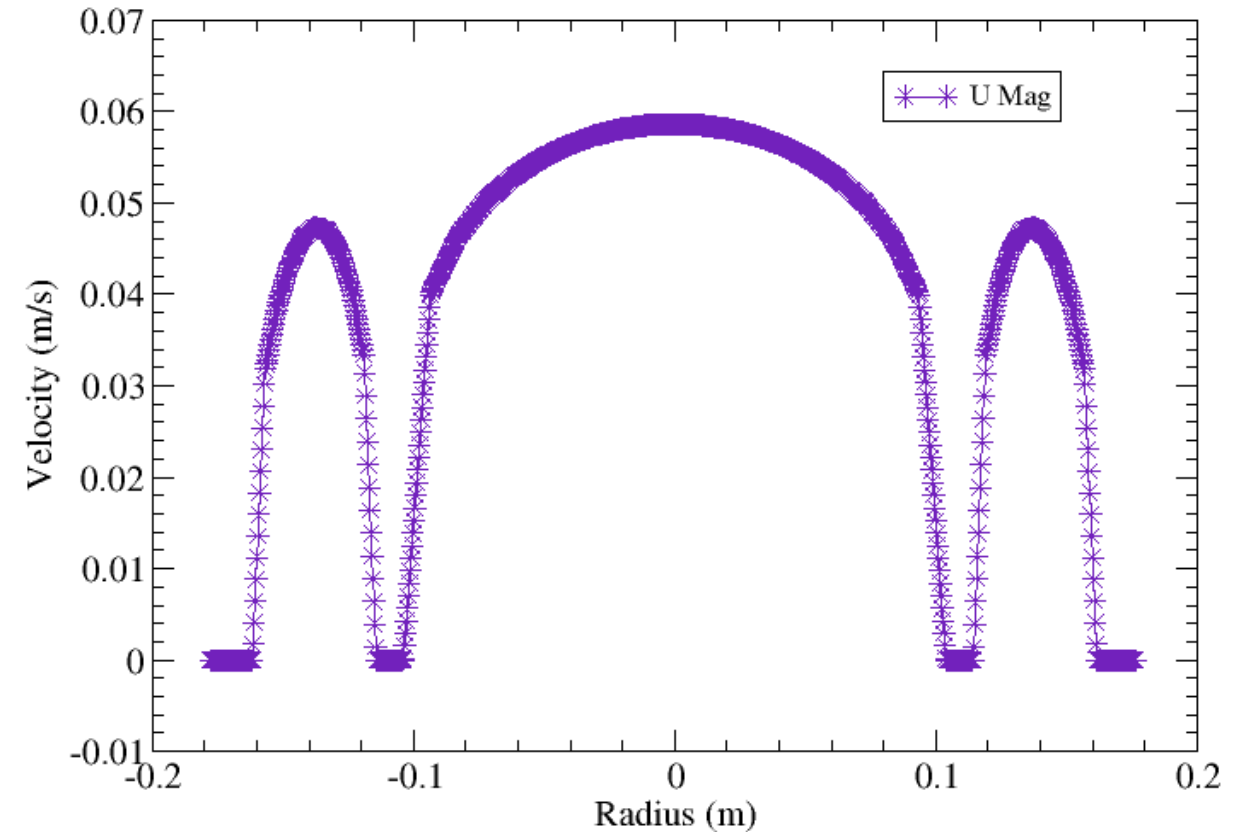


Laminar regime



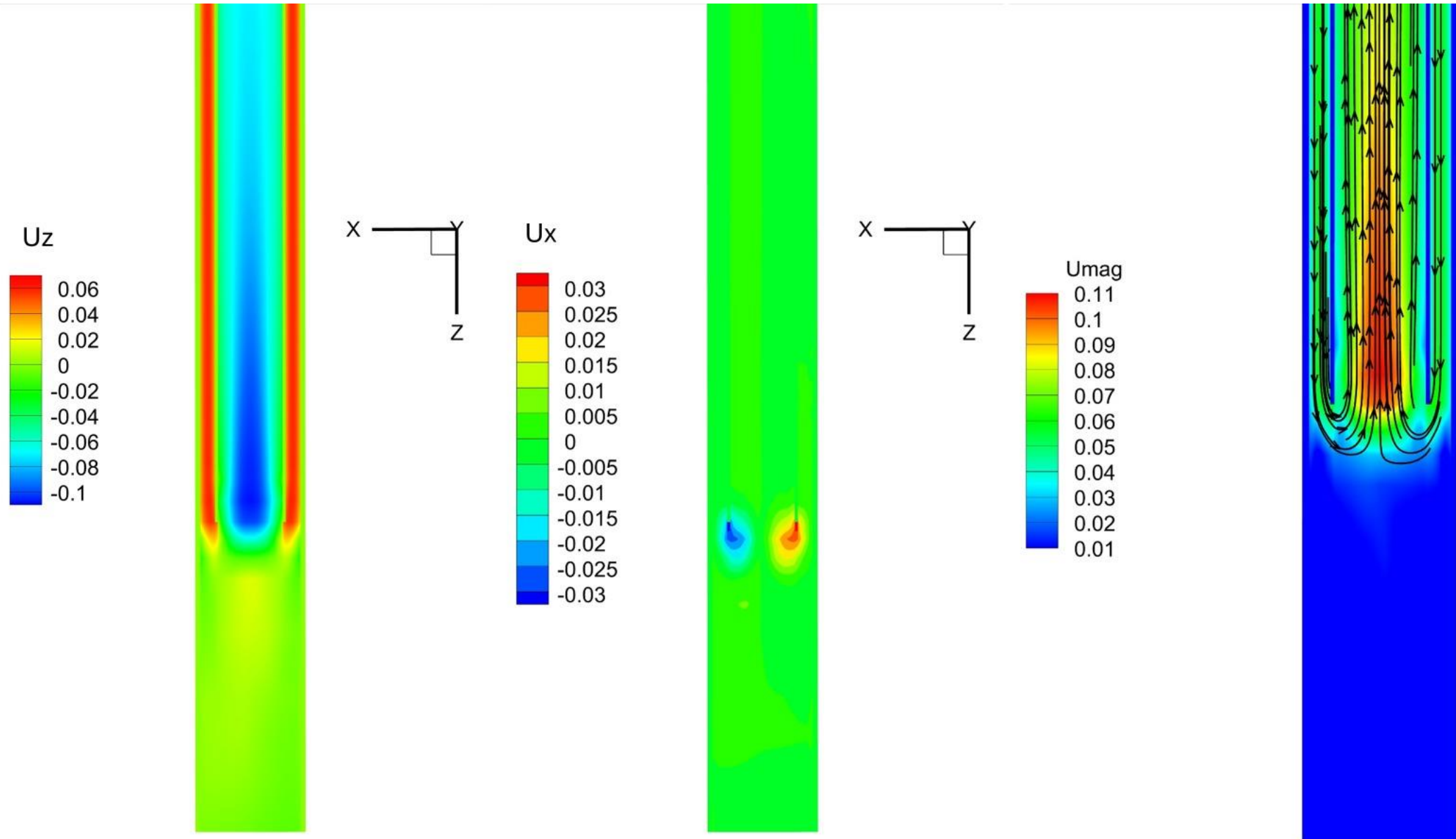
$z = 25m$

Turbulent regime



$z = 25m$

Bottom hole region:



Model Validation for Laminar flow:



Constant wall temperature condition (Insulated pipe):

By simulation: $V_{avg} = \frac{2}{R^2} \int_0^R u(r, z) r dr$

By theory: (Frank P. Incropera et al., 2017)

$$T_m = \frac{2}{V_{avg} R^2} \int_0^R T(r, z) u(r, z) r dr$$

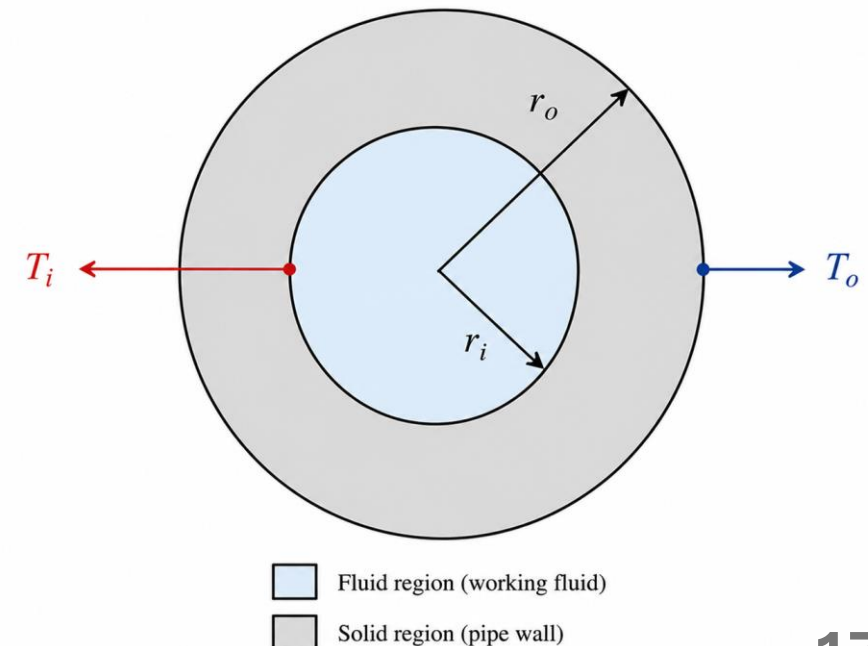
$$h = \frac{q_s}{T_s - T_m} = 19.3621 \frac{W}{m^2 K}$$

$$q_s = \frac{k(T_o - T_i)}{r_o \ln\left(\frac{r_o}{r_i}\right)}$$

$$Nu = \frac{h D}{k_{fluid}} = 6.7044$$

$$Nu = 3.66$$

Plane view of cross-section of pipe with thickness



Model Validation for Laminar flow:



Constant wall heat flux condition (Conductor pipe):

By simulation: $V_{avg} = \frac{2}{Ri^2 - r_o^2} \int_{r_o}^{Ri} u(r, z) r dr$

By theory: (Frank P. Incropera et al., 2017)

$$T_m = \frac{2}{V_{avg}(Ri^2 - r_o^2)} \int_{r_o}^{Ri} T(r, z) u(r, z) r dr$$

$$Nu = 4.86$$

$$h = \frac{q_s * A_o}{(T_s - T_m) A_i} = 8.63809 \frac{W}{m^2 K}$$

$$q_s = \frac{k(T_o - T_i)}{R_o \ln\left(\frac{R_o}{R_i}\right)}$$

$$Nu = \frac{h D_h}{k_{fluid}} = 3.2956$$

Turbulent Flow Validation using Dittus-Boelter Correlation:



By simulation:

$$V_{avg} = \frac{2}{R^2} \int_0^R u(r, z) r dr$$

$$T_m = \frac{2}{V_{avg} R^2} \int_0^R T(r, z) u(r, z) r dr$$

$$h = \frac{q_s}{T_s - T_m} = 262.103 \frac{W}{m^2 K}$$

$$q_s = \frac{k(T_o - T_i)}{r_o \ln\left(\frac{r_o}{r_i}\right)}$$

$$Nu = \frac{h D}{k_{fluid}} = 90.75$$

By theory: (Frank P. Incropera et al., 2017)

$$Nu: 0.023 \cdot Re^{0.8} \cdot Pr^{0.333}$$

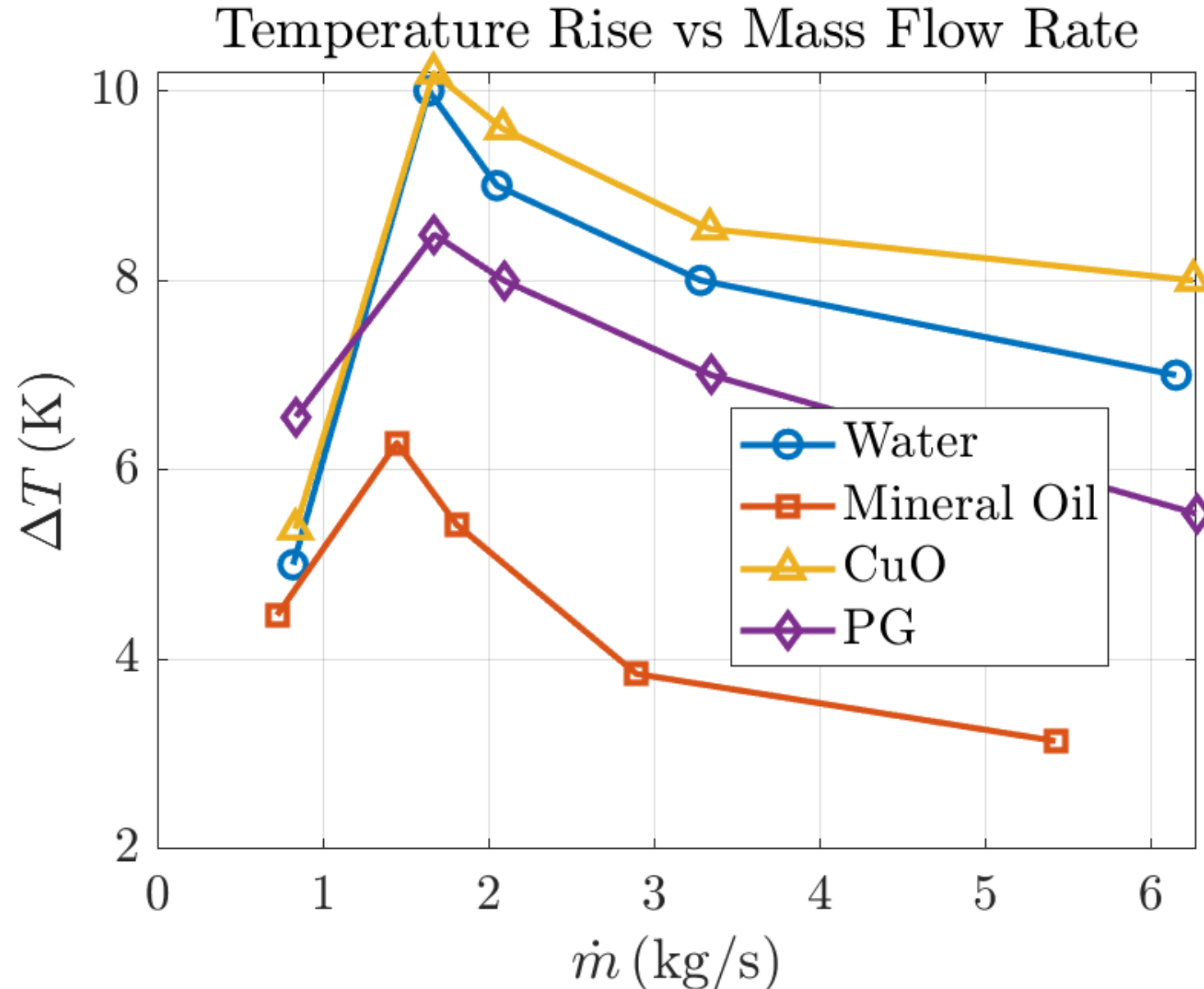
$$Nu: 83.57$$

Parametric study:

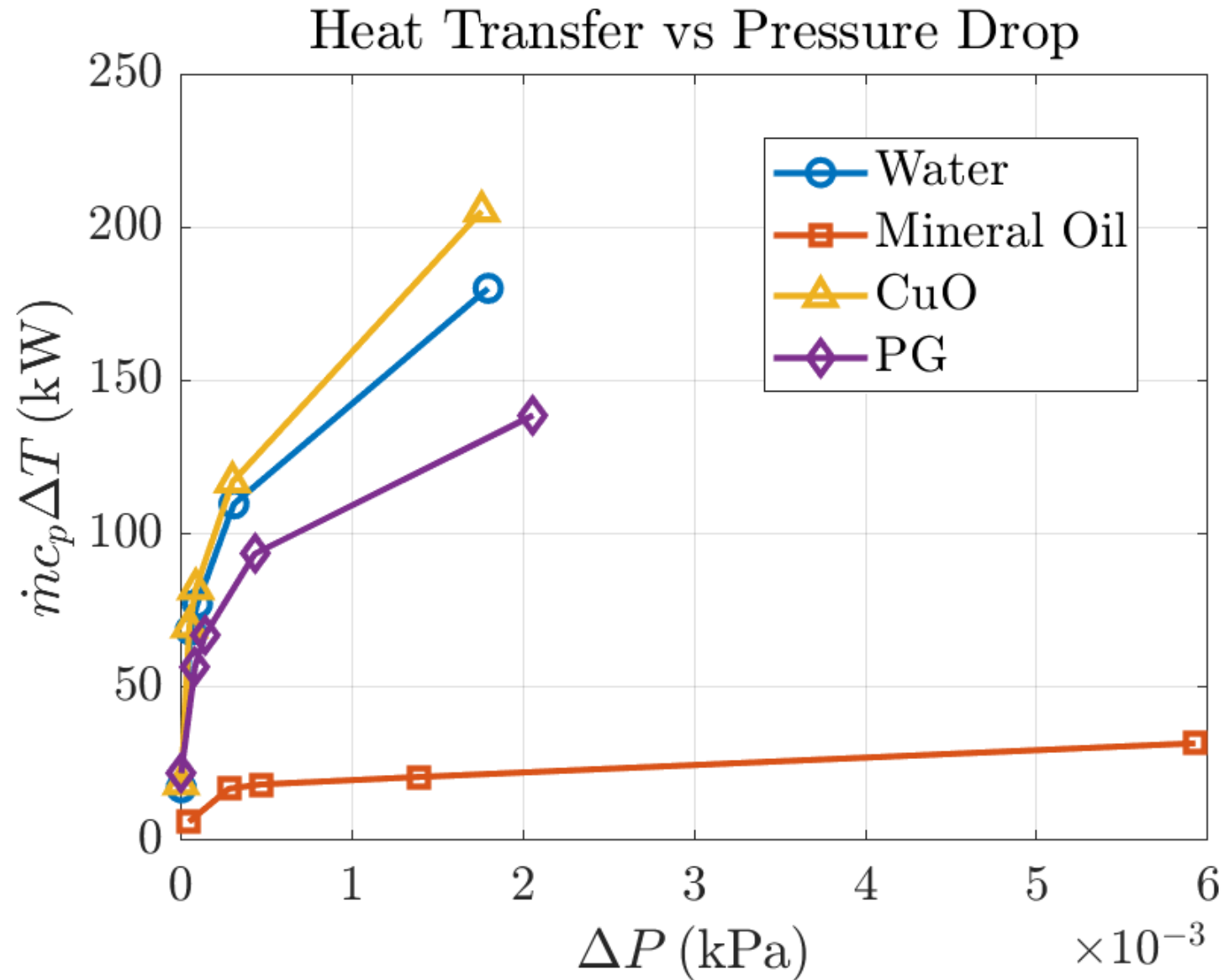


Working Fluids	Thermal Conductivity(k)	Specific Heat(C_p)	Viscosity(μ)	Density(ρ)
Water	High	4181	0.001	998.2
CuO/water (R.Kavitha et al. 2022)	High(enhanced)	4100	0.00084	1015
Mineral Oil (PubChem)	Low	1860	0.00989	880
PG (DowCAL Geo-200)	Moderate	3990	0.00174	1018

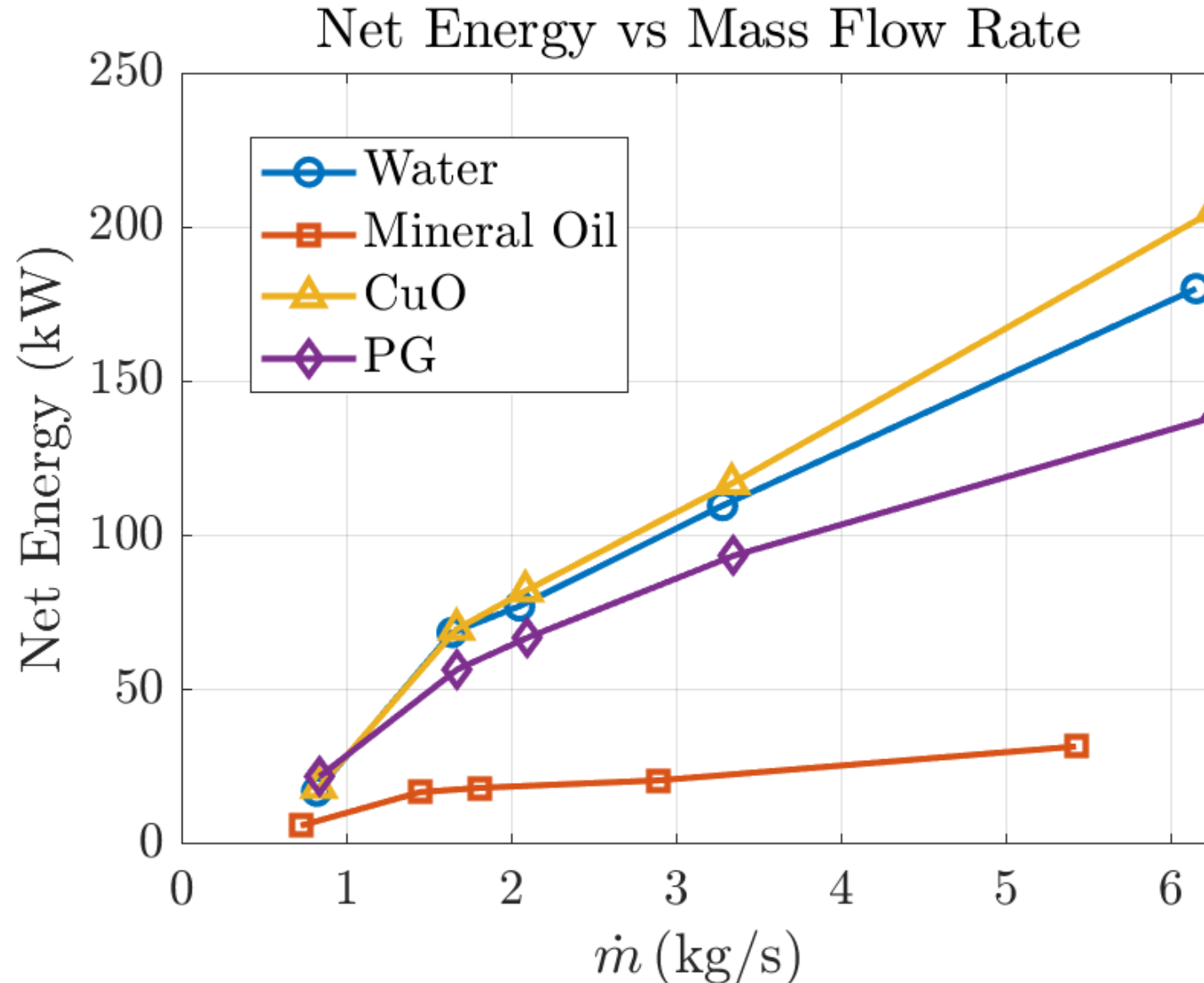
Effect of Mass Flow Rate on ΔT :



Effect of Heat Transfer rate on ΔP :



Effect of Net Energy on Mass Flow Rate:



$$Q_{net} = Q_{th} - W_p$$

Q_{th} = Total thermal gain

W_p = Hydrological power

$$W_p = \frac{m \cdot \Delta P}{\rho}$$

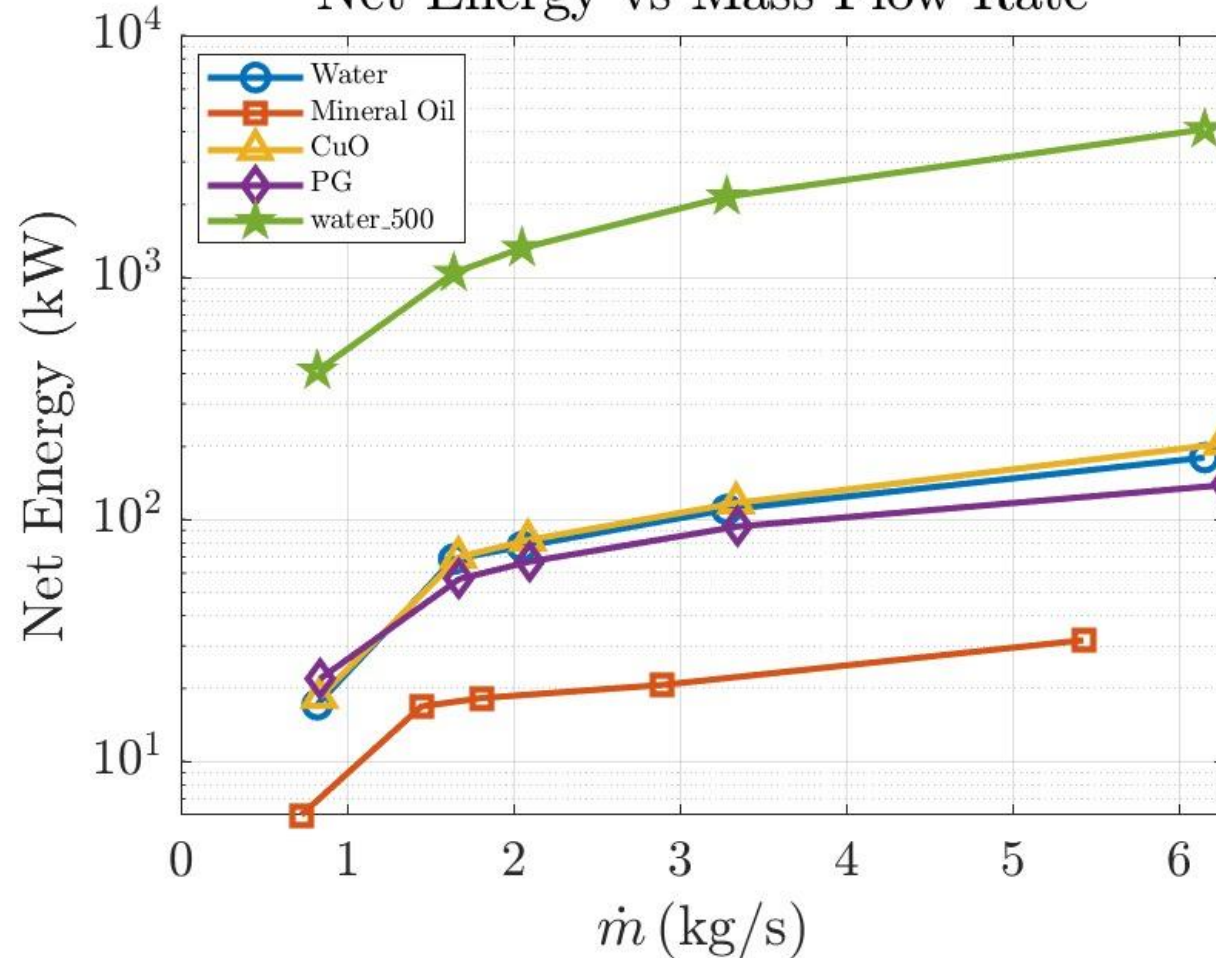
Scaling Analysis: (50 vs 500 meters)

CHEMICAL ENGINEERING

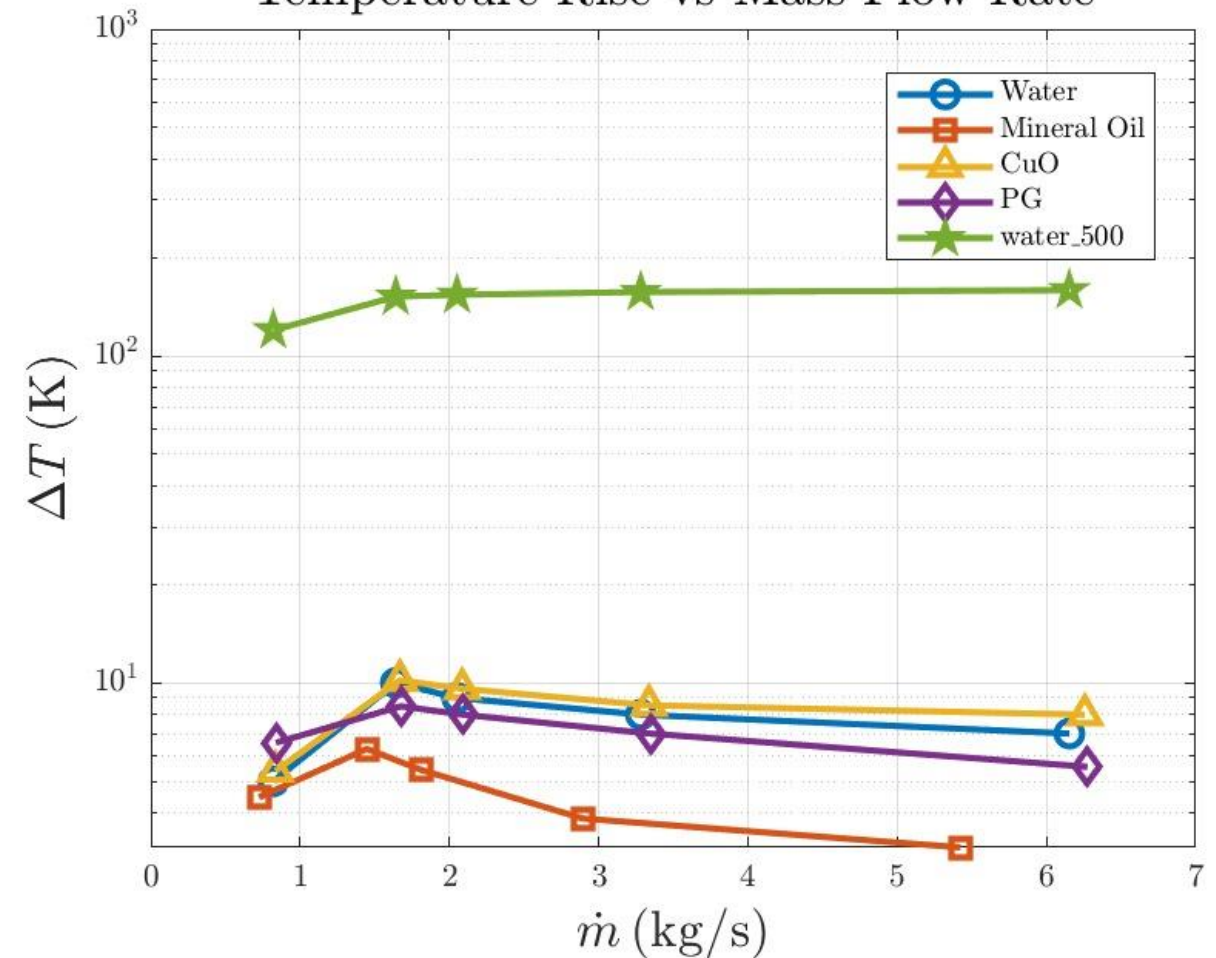
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Net Energy vs Mass Flow Rate



Temperature Rise vs Mass Flow Rate



Conclusion:



- Successfully **developed** a 3D, steady state, **finite volume-based** computational model, in OpenFOAM to solve **conjugate heat transfer and fluid flow phenomena**.
- Turbulent flow gives **high heat extraction** than laminar flow
- **CuO/water nanofluid** achieved the highest net energy.
- **Scaling** the system to 500 meter delivered **22-fold higher net energy** than 50 meter system.



Thank you everyone

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